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$$0.3 \frac{1}{\min} \left(\frac{\text{dm}^3}{0.2 \text{ mol}} \right) = 0.25 \frac{\text{dm}^3}{\text{mol} \cdot \min}$$

CHEMICAL REACTION ENGINEERING

MIDTERM EXAM 2

CHE 321, SPRING 2026, CLOSED BOOK, TIME: 75 MIN

Consider the following liquid phase reactions:

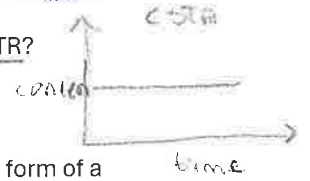


$$S_{B/X} = \frac{r_B}{r_X}$$

$$0.3 \frac{1}{\min} \left(0.2 \frac{\text{mol}}{\text{dm}^3} \right) = 0.004 \left(\frac{\text{mol}}{\text{dm}^3} \right)^{1/2}$$

$$V = F_{A0} \frac{X}{-r_A} \quad F_{A0} = F_{A0}(1-X)$$

The reactions are carried out isothermally first in a continuously stirred tank reactor (CSTR) to maximize the overall selectivity of the desired product B over the undesired ones X and Y. The outlet of the CSTR is then fed to an isothermal plug flow reactor (PFR) to increase the conversion of species A to 99%. Species A is fed to the first reactor at a concentration of $0.2 \frac{\text{mol}}{\text{dm}^3}$ in a liquid stream with a volumetric flow rate of $10 \frac{\text{dm}^3}{\text{min}}$.



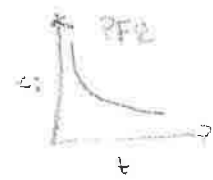
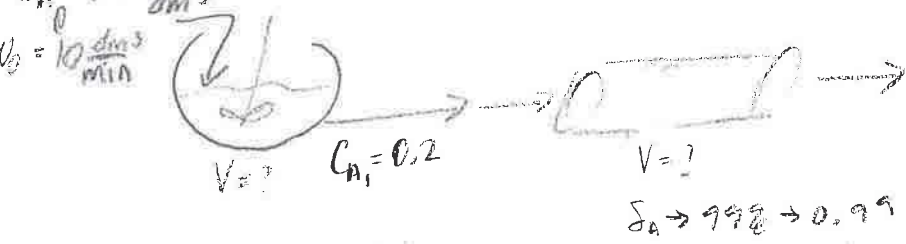
1- Calculate the volume of the CSTR. What is the overall selectivity and conversion in the CSTR?

2- Calculate the volume of the PFR.

Note: for part 2, if your calculator does not evaluate definite integrals, leave your answer in the form of a definite integral.

$$C_{A0} = C_A \text{ for CSTR}$$

$$F_{A0} = 2 \frac{\text{mol}}{\min} \quad C_{A0} = 0.2 \frac{\text{mol}}{\text{dm}^3} \quad v_0 = 10 \frac{\text{dm}^3}{\min}$$



1)

$$S_{B/X} = \frac{k_2 C_A}{k_1 C_A^{1/2}} \rightarrow \frac{k_2}{k_1} C_A^{1/2} \rightarrow \frac{0.3 \text{ min}^{-1}}{0.004 \left(\frac{\text{mol}}{\text{dm}^3} \right)^{1/2}} \times 0.2 \frac{\text{mol}}{\text{dm}^3} \rightarrow 15$$

$$S_{B/Y} = \frac{k_2 C_A}{k_3 C_A^2} \rightarrow \frac{k_2}{k_3} C_A^{-1} \rightarrow \frac{0.3 \text{ min}^{-1}}{0.25 \frac{\text{dm}^3}{\text{mol}} \cdot \text{min}^{-1}} \left(0.2 \frac{\text{mol}}{\text{dm}^3} \right)^{-1} \rightarrow 16$$

21 is overall selectivity?

$$V = \frac{F_{A0} \cdot X}{-r_A} \rightarrow \frac{F_{A0} \cdot X}{-k_2 C_A} \rightarrow V = \frac{2 \frac{\text{mol}}{\min} \cdot X}{-0.004 \frac{\text{mol}}{\text{dm}^3} \cdot \text{min}^{-1}} \rightarrow 33.3 X \text{ dm}^3$$

$$0.3 \frac{1}{\min} \left(0.2 \frac{\text{mol}}{\text{dm}^3} \right) = 0.004$$

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assume $V = V_0 \frac{\text{dm}^3}{\text{min}}$

$$V = \frac{F_{A0} \cdot X}{k_2 C_A} \rightarrow \frac{2 \frac{\text{mol}}{\text{min}} \cdot X}{0.006 \frac{\text{mol}}{\text{min} \cdot \text{dm}^3}} \rightarrow 33.3X \text{ dm}^3 = 10 \frac{\text{dm}^3}{\text{min}} \cdot \frac{1}{3} \frac{\text{min}}{\text{dm}^3} X$$

$$1011 = X$$

I don't know how to solve to get conversion or regular vol with the variables we were given at the beginning

I think I need to find

How to find the vol for CSTR I don't know how to relate selectivity to everything this is the best I got

$$V = \frac{2 \text{ mol/min} (1.11)}{0.006 \frac{\text{mol}}{\text{min} \cdot \text{dm}^3}} = 37 \text{ dm}^3$$

$$2) \quad V = F_{A0} \int_0^X \frac{dx}{-r_A}$$